Predicting Star-Level Contracts in the NBA

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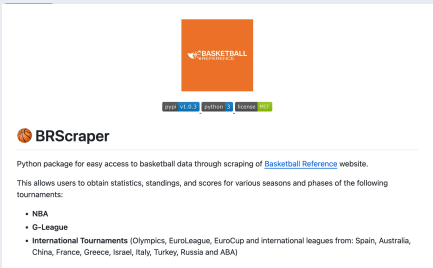
Research Question:

Which early career performance factors lead to a "star-level" contract during a player's career?

- Hundreds of advanced stats and metrics exist to measure different aspects of player performance
- Can we use these to predict how high a player's contract value will peak?

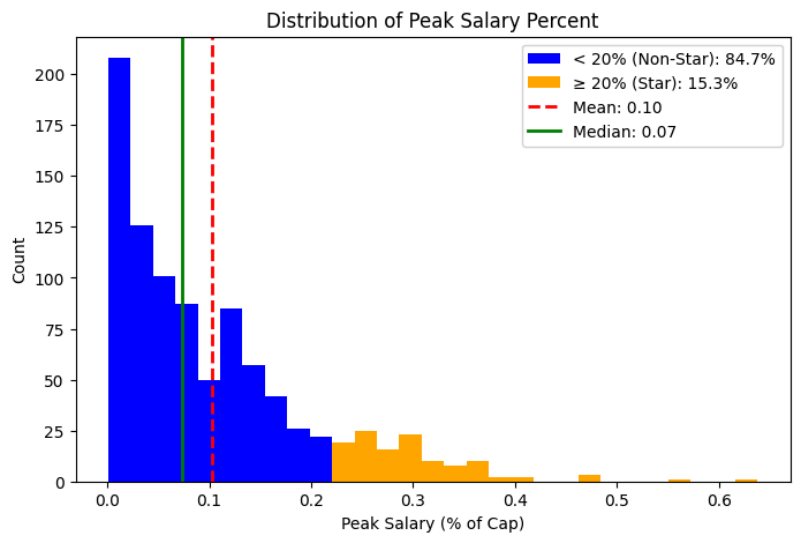
Data



- Advanced stats scraped from BasketballReference using Gabriel Pastorello's BRScraper tool
- Contract data scraped from BasketballReference by scraping every team-season page to extract player salaries, matching them to the list of players, and calculating each player's max salary
- 924 players drafted between 1991 and 2015 with ≥2 seasons played included
- Year 1 and Year 2 stats:
  - Advanced stats (TS%, eFG%, USG%, PER, VORP, BPM, WS, WS/48, TRB%, AST%, STL%, BLK%)
  - Per game stats (Points, Total Rebounds, Assists, Steals, Blocks)

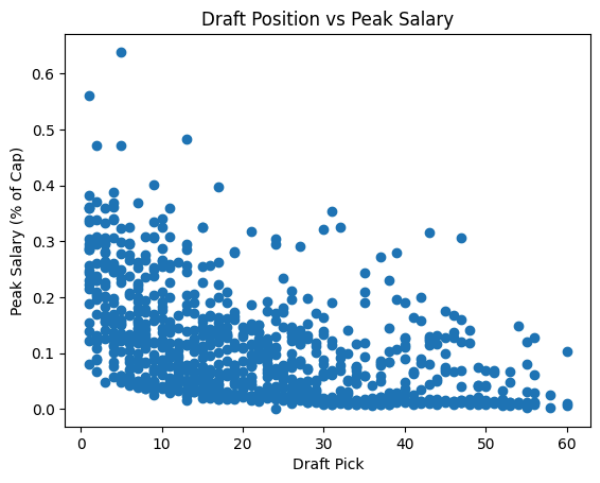
Salaries

- Player salaries represented as % of that season's salary cap to account for cap inflation
- Mean: 10% of the salary cap or ~$15.5 million under the current salary cap of $154.64 million
- Median: 7% of cap or ~$10.8 million currently
- We defined a "star-level" contract as one that is greater than 20% of the salary cap for the given year.



Draft distribution

- As expected, the number of players who peaked with a "star contract" is higher at the beginning of the draft
- Two main reasons:
  - Players drafted higher tend to be better
  - Players drafted higher that don't perform as well tend to be given more time to develop, as the team's "commitment" to them is larger.

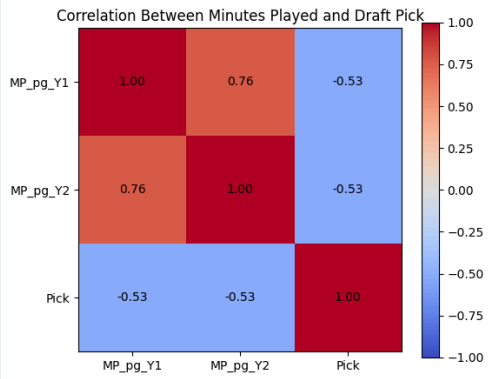


Playing Time

Moderate negative correlation between draft position and early-career playing time(r ≈ -0.53):

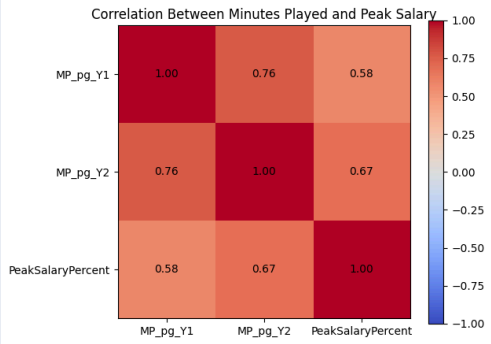
- Negative values because lower draft pick = better
- Higher draft picks receive more minutes to start

- Higher expectations earlier on and increased opportunity on weaker teams



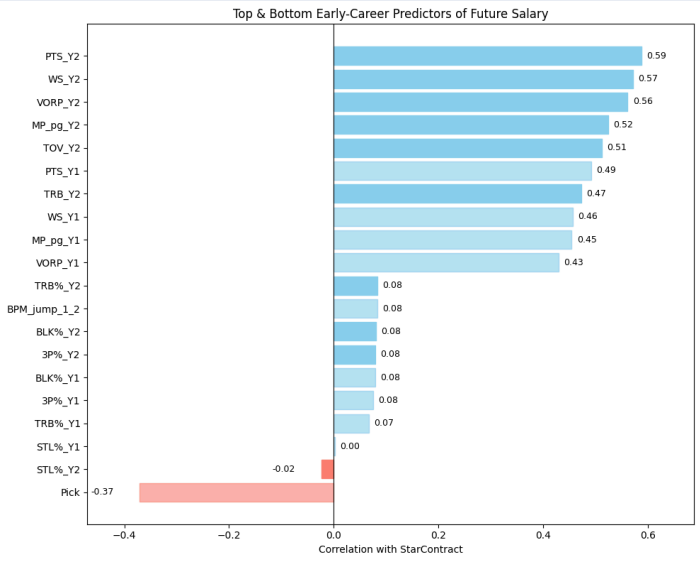
Stronger positive correlation between playing time and peak salary percentage

- This is expected as players who make an immediate impact and play more in their earlier seasons tend to peak higher.
- The increase between Year 1 and Year 2 is also expected as players who have a consistent role in their first season usually receive increased playing time in their second season.



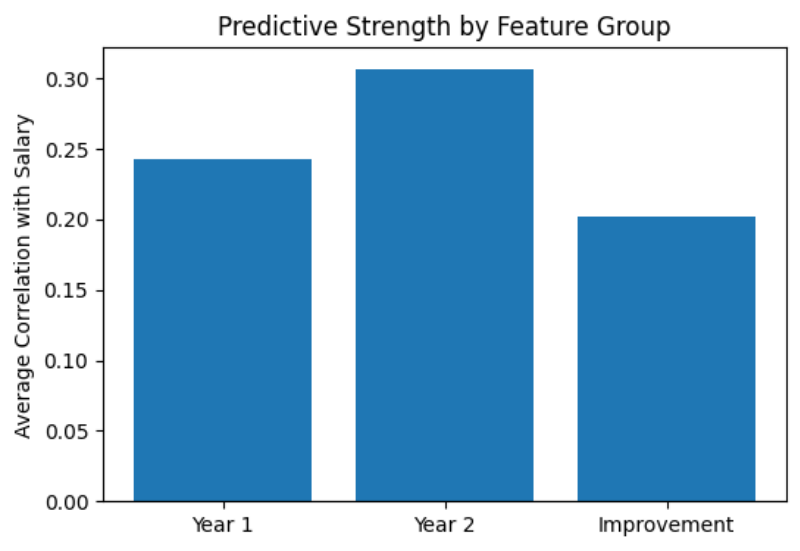
Early Production and Role Drive Future Earnings

- Year 2 Production and opportunity metrics most important
  - Points, Win Shares, VORP, Minutes (r ≈ 0.52–0.59)
- Year 2 performance is more predictive than Year 1
- Efficiency metrics show weaker relationships
  - 3P% very low, eFG%, TS%, FT% all towards the middle
- Improvement metrics not showing much impact so far
- Draft position somewhat of a factor (r ≈ -0.37)



Year 1 vs Year 2 vs Improvement

- Year 2 metrics are the most predictive
  - Average correlation: 0.31 vs 0.24 (Year 1) and 0.20 (Improvement)
- Year 1 performance still matters, but not as strong of a correlation
- Improvement metrics have limited predictive power
  - Growth is less important than overall production level



Modeling

- Lasso/Ridge/PCA: R^2 of .47
- Classification Tree: R^2 of .886
- Gradient Boosting: R^2 of .891
- Random Forest: R^2 of .897
- Most influential variables across models: PTS_Y2, PER_Y2, MP_pg_Y2, VORP_Y2
- Initial models not very accurate, but later use of decision trees had a far higher accuracy, with Year 2 variables showing up frequently across all of those models

Where the model fails

Some players achieve star-level contracts despite very low early-career Win Shares, revealing important limitations in models based only on early performance.

What the data shows

- Several players with low average WS in first two seasons still reached high-value careers
- Examples include late developers and role expanders
- Early production does not fully capture long-term player value

Example players (Star Contracts with Low Early WS)

Player	DraftYear	WS_Y1	WS_Y2	avg_WS_first2	PeakSalaryPercent
D'Angelo Russell	2015	0.0	1.3	0.65	0.23
Julius Randle	2014	-0.1	1.6	0.75	0.20
Zach LaVine	2014	-0.7	2.6	0.95	0.28
Khris Middleton	2012	0.9	2.7	1.80	0.28
DeAndre Jordan	2008	1.3	1.5	1.40	0.21

Interpretation

Early-career Win Shares alone fail to capture:

- player development trajectories
- increased usage and opportunity
- late-blooming performance growth

These cases highlight systematic blind spots in purely early-performance-based salary prediction models.

- These 5 players are all at least 1x All-Stars and and two are champions
- All have played key roles for good teams, and if they had been given up on due to poor early career performance this wouldn't be the case.

Key Takeaways so far

- Performance more predictive than improvement
  - Year 2 metrics (production + impact) are stronger predictors than growth or improvement alone
- Production & opportunity matter most
  - Points, WS, VORP, PER, and minutes most strongly correlated with future star-level contracts
- Draft position still influential
  - Early picks get more opportunity and larger contracts, after some control for performance
- Efficiency less predictive early
  - TS%, eFG%, 3P%, FT% show weaker correlation with star-level contracts in first two years

Implications for Stakeholders

Primary Audience: NBA front offices

Early identification of future stars

- Helps teams recognize players on star trajectories before contract decisions
- Can help identify undervalued players

Contributes to rookie extensions and longer-term financial planning

- Predicting future player breakouts can help with salary planning in an era of the NBA where penalties for exceeding the salary cap can be harsh
- Can reduce likelihood of overpaying players without ”star-level” indicators

Ethical/Societal Implications

Oversimplification of talent evaluation

- Important traits like leadership, locker room value, and injuries aren't captured in stats

Risk of missing out on late bloomers

- Early improvement doesn't guarantee later career success, many players are late bloomers and this data only covers a player's first two years.

Potential for bias

- Historical biased contract trends might affect how the model treats certain players
- Higher drafted players getting larger contracts, lack of appreciation for defensive specialists and stretch bigs.